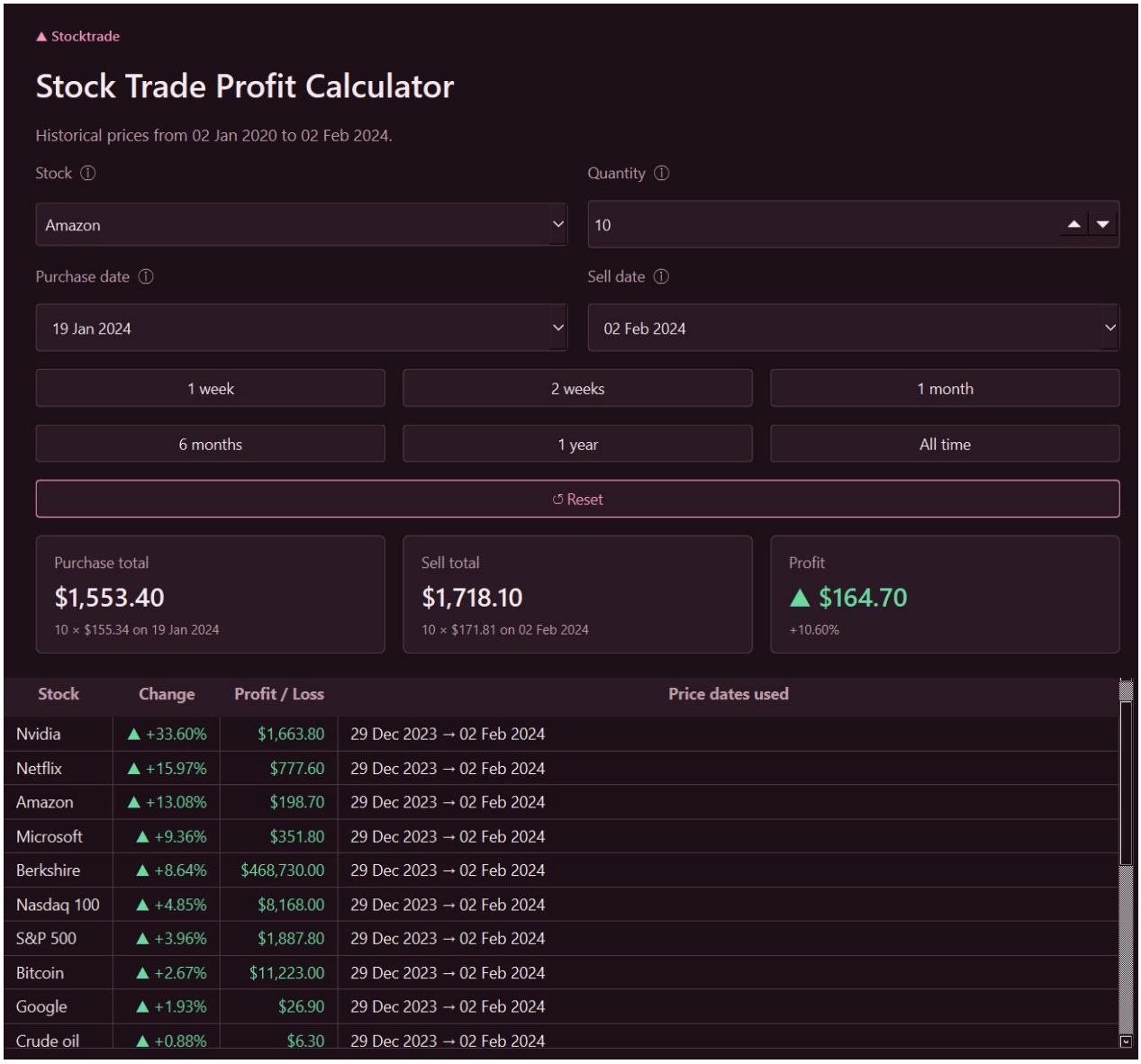
BSC – HGP - Assignment 1

Stock Trade Profit Calculator app

UI Design Document

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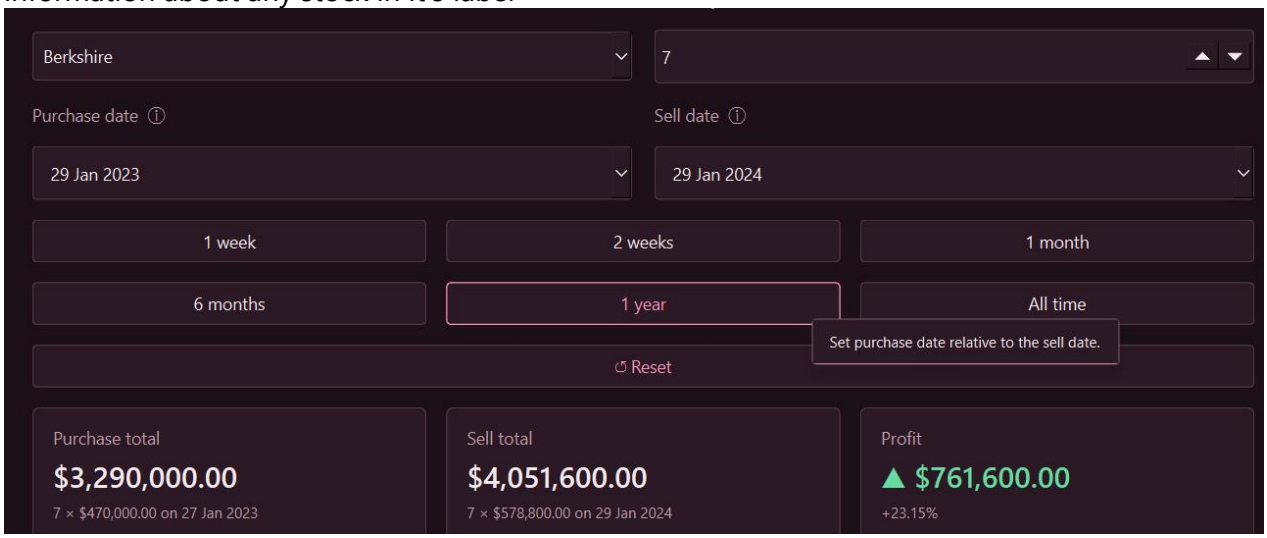
1.



The overview of the app before any user input.



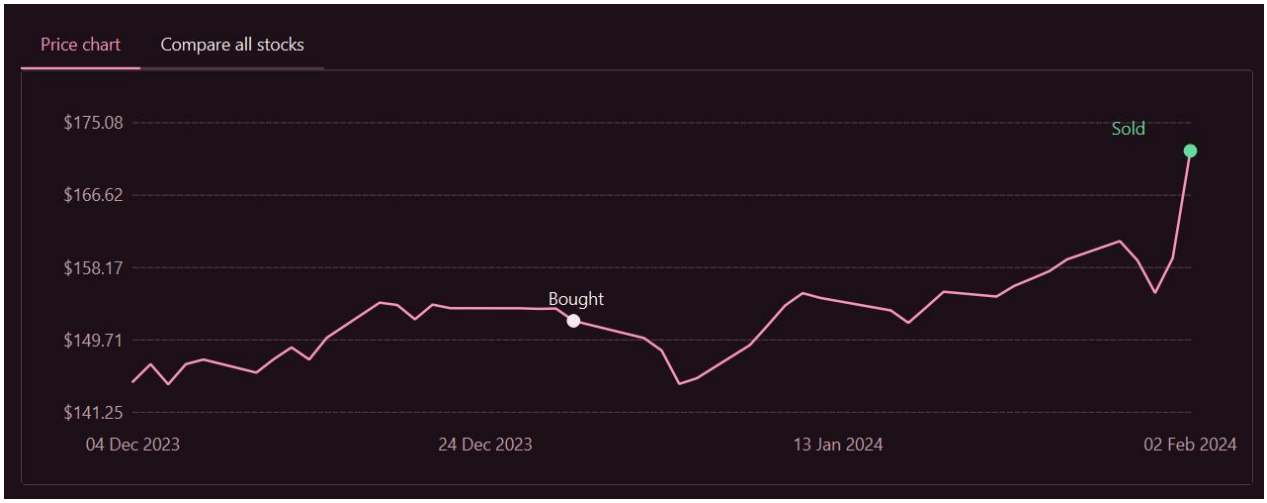
The overview of the interactive price chart. Hovering on any part of it would give some information about any stock in it's label



The stock, it's quantity, buy date, and sell date have been changed by the user. The sell date can only be set after the purchase date has been set. Although if any of the pre written durations have been chosen then the purchase date is set relative to the sell date.
The metrics of cost (Profit, Sell and buy total) are calculated immediately.

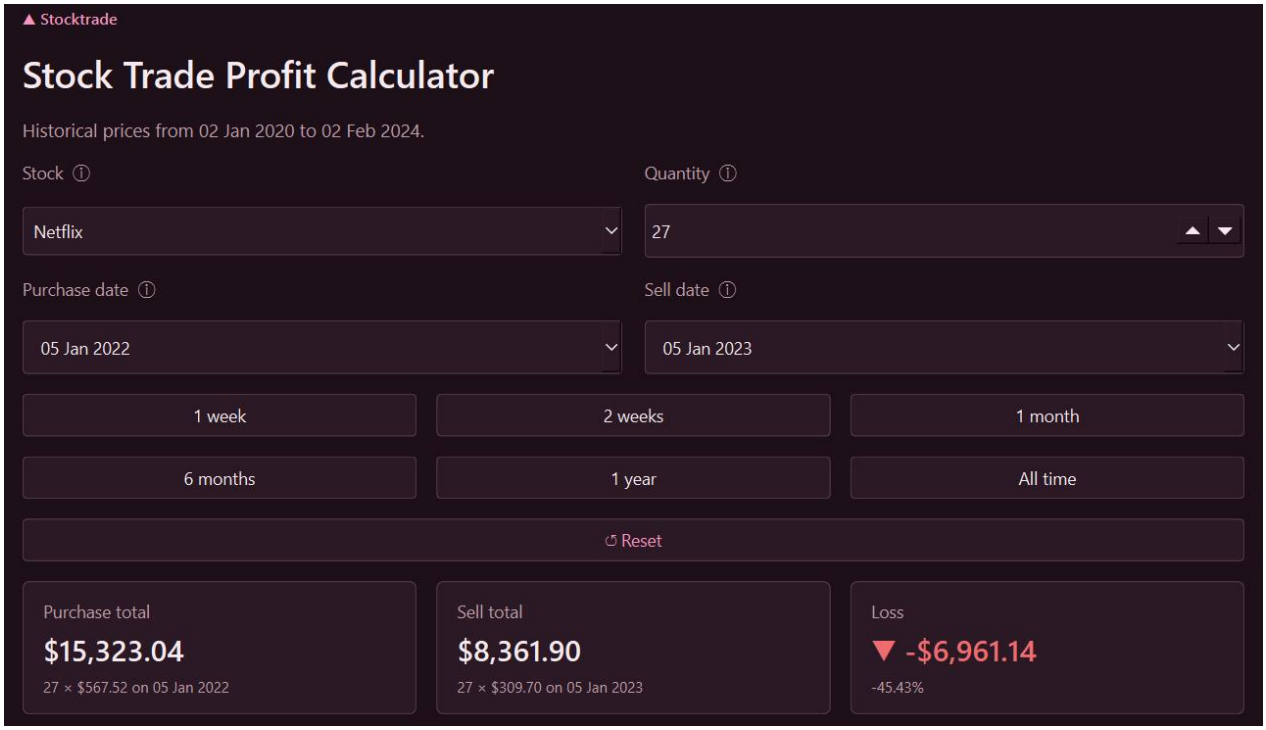
Stock	Change	Profit / Loss	Price dates used
Nvidia	▲ +33.60%	$1,663.80	29 Dec 2023 → 02 Feb 2024
Netflix	▲ +15.97%	$777.60	29 Dec 2023 → 02 Feb 2024
Amazon	▲ +13.08%	$198.70	29 Dec 2023 → 02 Feb 2024
Microsoft	▲ +9.36%	$351.80	29 Dec 2023 → 02 Feb 2024
Berkshire	▲ +8.64%	$468,730.00	29 Dec 2023 → 02 Feb 2024
Nasdaq 100	▲ +4.85%	$8,168.00	29 Dec 2023 → 02 Feb 2024
S&P 500	▲ +3.96%	$1,887.80	29 Dec 2023 → 02 Feb 2024
Bitcoin	▲ +2.67%	$11,223.00	29 Dec 2023 → 02 Feb 2024
Google	▲ +1.93%	$26.90	29 Dec 2023 → 02 Feb 2024
Crude oil	▲ +0.88%	$6.30	29 Dec 2023 → 02 Feb 2024

Stock Comparison tab

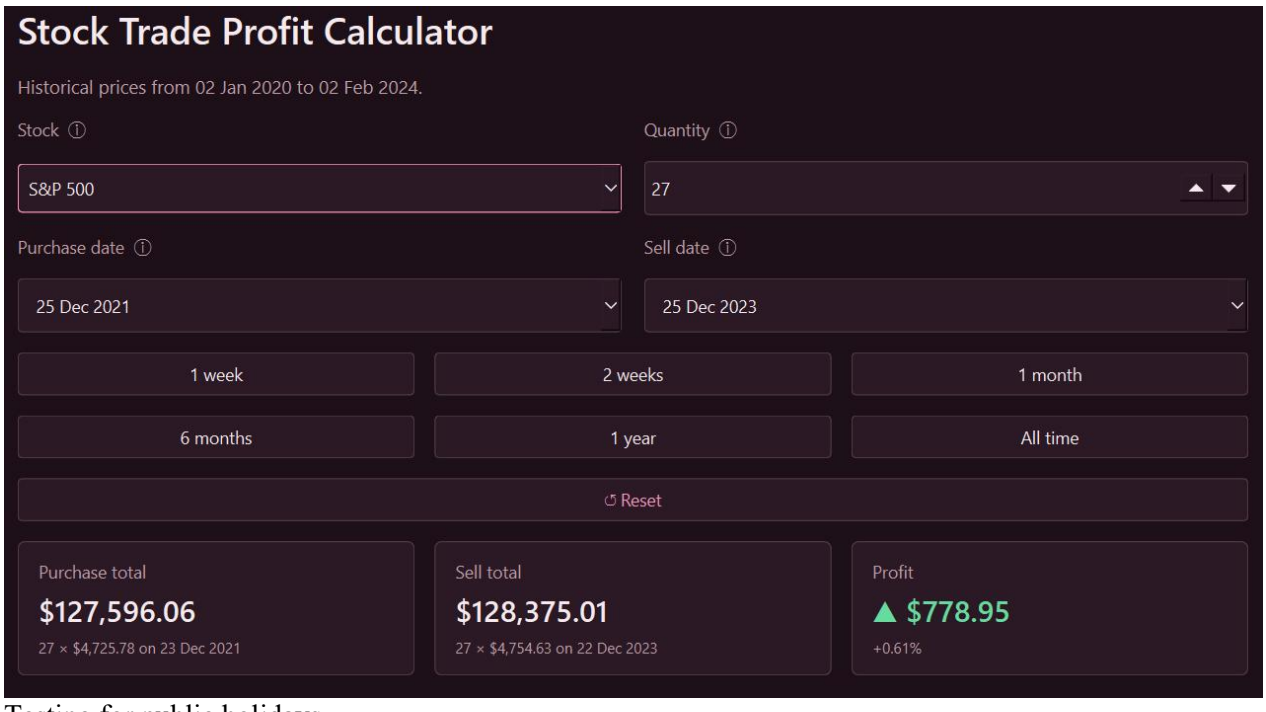


The interactive price chart is also immediately updated **(Additional Feature)**





Comparison to stocks operating on a loss.



Testing for public holidays



2. Components of my app that working properly:

Trade Input and Stock Assets selection (Stock, buy/sell date, quantity and duration)
Reset button for restoring the default stock assets
(Additional Features) Financial Summary: Price Interaction visualization with purchase and sale markers. Stock comparison analysis that ranks the different stocks and take the dates into consideration
Error handling & Edge case guidance: Error messages, tool tips and unavailable date selection
Responsive layout with windows re sizing
Instant Recalculation and results cards (profit /loss, purchase total and sell total)
Parsing and caching of all the data in the csv file

Components that not performing well:
Fonts are quite basic due overall theme and usability principles
Unavailable dates are to be visually dimmed but it causes a clash due to the overall dark theme of the app.

3.

Stock Asset Selection

Control/Input Type:

- *Stock Selector:* A drop down combo box (QComboBox) was used because the data set contains a finite list of stock details and assets. Drop down menus prevent spelling errors, eliminate invalid inputs, and they adhere to Nielsen's heuristic of recognition.
- *Purchase and Sell Dates:* Drop down calendar date pickers (QDateEdit) was used to provide a visual navigation (month after months) while enforcing strict rules (Must be between 2 January 2020 and 1 February 2024; The sell date can only be picked after the purchase date has been picked; If the user uses one of the duration, then the purchase date is relative to the sell date).
- *Quantity:* A dedicated integer spin box input (QSpinBox) was used to ensure that negative shares, fractional purchases and non integer were forbidden. Used with explicit minimum (1) and step (1) increments

Location:

- Positioned at the left-hand side of the card at the top of the visual reading hierarchy (following the Western style of reading top-to-bottom, and left-to-right reading). The trade inputs are3 group together to establish a difference between the results and show an intuitive setup.

Colour:

- Backgrounds --> deep muted plum/charcoal (#1D1217 / #23161C); Border -->subtle mauve (#4A2E3B); Canvas --->#140D10 .
- In active states -->bright rose-gold highlight (#F28B82 / #FFB4AB),Text --->off-white (#FCEEF2).

Size:

- Input heights are standardized to 40px with 12px internal padding to meet touch-target accessibility standards and ensure clear desktop cursor clickability without feeling cramped.

Style Decisions:

- Subtle rounded corners softens the technical aesthetic. Labels sit directly above inputs in semi-bold uppercase with informational tooltip icons (i) to communication input details.

Duration Shortcuts and Reset to Default Button

Control/Input Type:

- Duration Shortcuts: A horizontal segments group hold the durations (1 week, 2 weeks, 1 month, 6 months, 1 year, All time). The user can easily access stock information.

- Reset Button: Restore the default stock trade and it's assets (Amazon, 10 shares, 2 weeks holding period to the latest date).

Location:

- Duration Shortcuts: Positioned below the date selectors inside the input panel, uses a direct spatial proximity (Gestalt law of proximity) between date inputs and date shortcuts.
- Reset Button: Placed at the bottom-right corner of the panel, ensures accessibility for quickly going back to the default trade without competing with other data controls.

Colour:

- Duration shortcut buttons--->dark plum background (#23161C) with soft rose borders. Hovering/active--->rose tint (#4A2E3B). Text----> (#FFFFFF).
- The Reset button ----->outline style with rose accents (#BAA1AA) to indicate a secondary action.

Size:

- Shortcut buttons are compact using legible 12px text, allowing duration labels to align cleanly on a single row without wrapping.

Style Decisions:

- Full words are used to prevent user confusion. Button borders have a 6px pill radius, differentiating quick shortcuts from input boxes.

Trade Results & Financial Summary (Purchase Total, Sell Total, Profit/Loss Cards)

Control/Input Type:

- Read-only KPI's to display metric on the cards. Output nodes for showing entered values, avoiding user confusion about typing the input.

Location:

- Placed across the top of the right-hand dashboard column, beside the input panel. Allows users to see the calculated results without scrolling.

Colour:

- Purchase and Sell Totals: Frame -->dark plum backgrounds (#1D1217)Text --->cream (#FCEEF2).
- Profit / Loss Card: Dynamically styled based on trade outcome:

Gain: Saturated emerald green (#4EBE88 text, #132B1E background tint).

Loss: High-contrast crimson red (#E25C6A text, #2F1418 background tint).

Dual Coding for Accessibility: Clear directional icons (▲ for profit, ▼ for loss) and text labels ("Profit" or "Loss") are used so colour-blind users do not have to use on color alone.

Size:

- Large cards spanning equal thirds across the results column. Numerical figures are shown in a bold font to establish visual hierarchy in the data outputs.

Style Decisions:

- Shown using fixed-width mono spaced typography so digits and decimal points remain strictly aligned during live recalculations. Sub-labels display the exact market dates used.

Price Interaction Visualisation (Interactive Price Chart with Buy/Sell Markers)

Control/Input Type:

- An integrated data visualization canvas (custom vector renderer via QPainter on desktop / HTML5 SVG on web). It transforms tabular historical daily closes into a smooth, continuous trendline with interactive pinpoint callouts.

Location:

- Centered directly beneath the KPI summary cards in the main results column, occupying the largest visual area on screen to support in-depth trend evaluation.

Colour:

- Background matches the deep plum card surface (#1D1217).
- The price trajectory is drawn with a luminous rose-gold stroke (#FFB4AB), flanked by a subtle translucent gradient fill beneath the line.
- Buy and sell transaction markers feature distinct high-contrast accent dots: a bright cyan/gold point for "Bought" and a vivid coral/rose point for "Sold", each accompanied by high-contrast pill badges and leader lines.

Size:

- Fixed minimum height of 280px with full width expansion across the card to ensure price fluctuations and horizontal spacing between trade dates are clearly legible.

Style Decisions:

- Includes clean horizontal price grid lines labeled in muted grey (#725562) at 11px. Markers feature direct textual pill labels ("Bought" and "Sold") with exact prices rather than an abstract legend, directly satisfying **Nielsen's heuristic of minimizing cognitive load.**

Stock Comparison Table (Ranked Performance & Asset Comparison)

Control/Input Type:

- An interactive, sortable tabular data grid (QTableWidget / semantic <table>) equipped with clickable row selectors. Each row acts as an active shortcut: clicking any row instantly selects that stock in the main calculator.

Location:

- Positioned in the lower section of the dashboard beneath the chart, functioning as a deeper exploratory tool once individual trade calculations are understood.

Colour:

- Header row features a darker plum surface (#160D12) with uppercase muted labels (#BAA1AA). Alternating rows utilize subtle contrast shading (#1D1217 vs #23161C) for clean horizontal scanning.
- Profit/Loss columns reuse the dual-coded emerald (#4EBE88) and crimson (#E25C6A) text with directional ▲/▼ indicators. Hovered rows illuminate with an accent border highlight.

Size:

- Displays all 19 assets in an organized, scrollable viewport with 36px row heights and padded cell margins, ensuring financial figures never truncate.

Style Decisions:

- Right-aligned numerical columns using monospaced figures guarantee instant vertical decimal alignment. A visual rank badge (#1 through #19) makes relative asset performance clear at a glance.

Error Handling, Validation & Edge Case Management

Control/Input Type:

- Inline warning containers, informational input tooltips (QToolTip), and input clamping.

Location:

- Informational error banners appear directly above or below the corresponding input fields, in line with the user's gaze.

Colour:

- Warning banners---->crimson background (#2F1418) Text: bright rose-red (#FF8993) Warning icon: amber warning icon (⚠).

Size:

- Compact height spanning the full width of the input card, providing enough surface area for text while preserving layout stability.

Style Decisions:

- **Error Prevention (Heuristic #5):** The sell date picker does not selecting dates till the purchase date has been selected
- **Non-Trading Days (Weekends/Holidays):** Rather than crashing or rejecting weekend dates, the system resolves to the nearest preceding market trading day and adds a note.

Responsive Layout with Window Re sizing

Control/Input Type:

- Layout container that easily adapts.

Location:

- Covers the entire application window canvas, managing spatial relationships across all panels.

Colour:

- Canvas backdrop ---->(#140D10), ensuring window borders, letterboxes, or re sized view ports maintain uniform visual continuity.

Size:

- Strict minimum window (e.g., 900px × 680px) to prevent component clipping, while dynamically re sizing chart coordinates and comparison table columns when expanded to full screen.

Style Decisions:

- On desktops, sizing policies (QSizePolicy.Expanding) allow the price chart and comparison table work with available window size. On narrower screens, the layout flows from a two-column side-by-side dashboard into a stacked single-column flow.

Instant Recalculation & Result Display

Control/Input Type:

- Event-driven reactive signals (valueChanged, currentIndexChanged, dateChanged connected directly to calculation methods).

Location:

- Runs along all the interactive input controls, triggeringa real-time repaint response across the output cards, chart markers, and comparison ranking.

Colour:

- Smooth QSS property transitions prevent visual flashes. When input values update, the text briefly highlights before settling into its final gain/loss state.

Size:

- Operates instantaneously with zero layout shift because mono spaced figures ensure consistent character widths regardless of number values.

Style Decisions:

- No traditional "Submit" button. Adhering to Nielsen's heuristic of Visibility of System Status, every user action produces immediate feedback.

Parsing and Caching of CSV Stock Data (Data Handling)

Control/Input Type:

- In-memory structured data store (hash-indexed dictionary).

Location:

- Within the application's core data layer, operating behind all UI widgets and stock comparisons.

Colour:

- Non-visual backend component; provides verified, clean data attributes to visual status indicators across the UI.

Size:

- Compact in-memory footprint (19 stock assets across 4 years of closing prices).

Style Decisions:

- All date parsing strictly uses and only validates (DD/MM/YYYY) format matching the source data in csv. Minimum to maximum boundaries, stock asset symbol lists, and trading calendars are always pre-cached on startup, enabling instant search without lagging